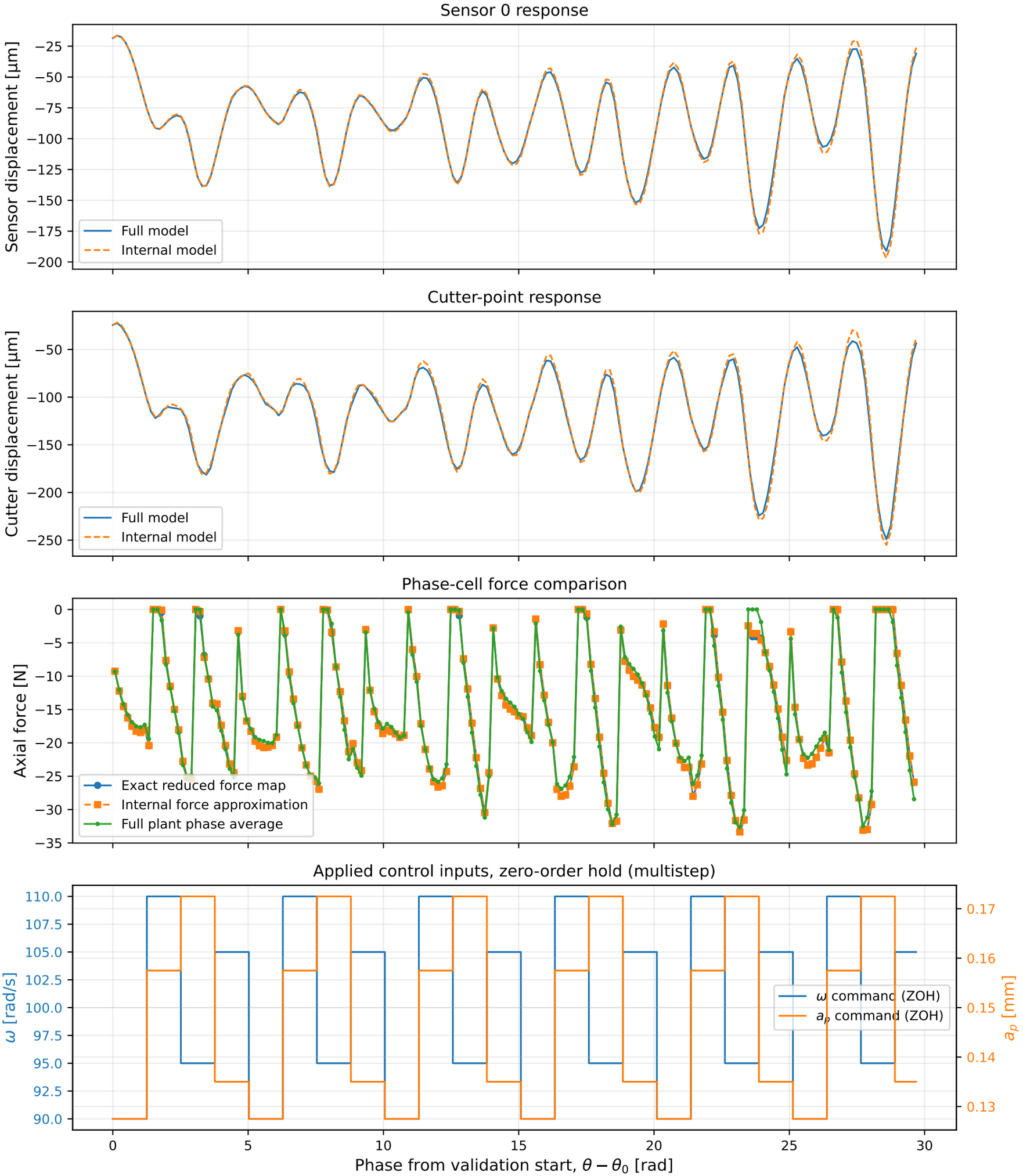
Sensor 0 response
Sensor displacement [µm]
−25
−50
−75
−100
−125
−150
−175
−200
Full model
Internal model
Cutter-point response
Cutter displacement [µm]
−50
−100
−150
−200
−250
Full model
Internal model
Phase-cell force comparison
Axial force [N]
0
−5
−10
−15
−20
−25
−30
−35
Exact reduced force map
Internal force approximation
Full plant phase average
Applied control inputs, zero-order hold (multistep)
ω [rad/s]
110.0
107.5
105.0
102.5
100.0
97.5
95.0
92.5
90.0
$a_p$ [mm]
0.17
0.16
0.15
0.14
0.13
ω command (ZOH)
$a_p$ command (ZOH)
0
5
10
15
20
25
30
Phase from validation start, $\theta - \theta_0$ [rad]